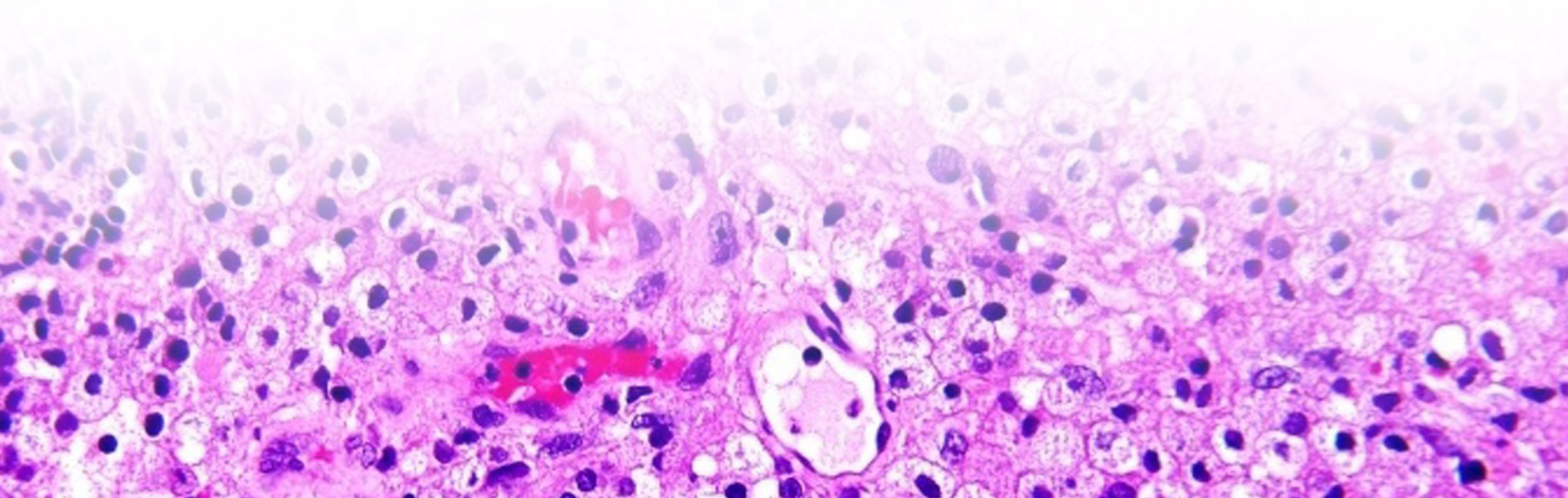
Student Notes

PATHOLOGY

1st Edition

Unit 25

CNS and EYE



Preface

Agam is a group of budding medicos, who are currently doing their under graduation in various Medical Colleges across Tamil Nadu and Pondicherry. The group was initiated on 18th November 2017, in the vision of uniting medicos for various social and professional causes.

We feel delighted to present you Agam Pathology notes prepared by Agam Divide and Rule 2020 Team to guide our fellow medicos to prepare for university examinations.

This is a reference work of 2017 batch medical students from various colleges. The team took effort to refer many books and make them into simple notes. We are not the authors of the following work. The images used in the documents are not copyrighted by us and is obtained from various sources.

Dear readers, we request you to use this material as a reference note, or revision note, or recall notes. Please do not learn the topics for the 1st time from this material, as this contain just the required points, for revision.

Acknowledgement

On behalf of the team, Agam would like to thank all the doctors who taught us Pathology. Agam would like to whole heartedly appreciate and thank everyone who contributed towards the making of this material. A special thanks to Vignesh M, who took the responsibility of leading the team. The following are the name list of the team who worked together, to bring out the material in good form.

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THE CENTRAL NERVOUS SYSTEM AND THE EYE

ESSAY:

1. Classify brain tumors. Glioma

SHORT NOTES:

1. Meningioma
2. TB meningitis
3. Examination of CSF
4. Prion disease
5. Acute Pyogenic Meningitis
6. Medulloblastoma

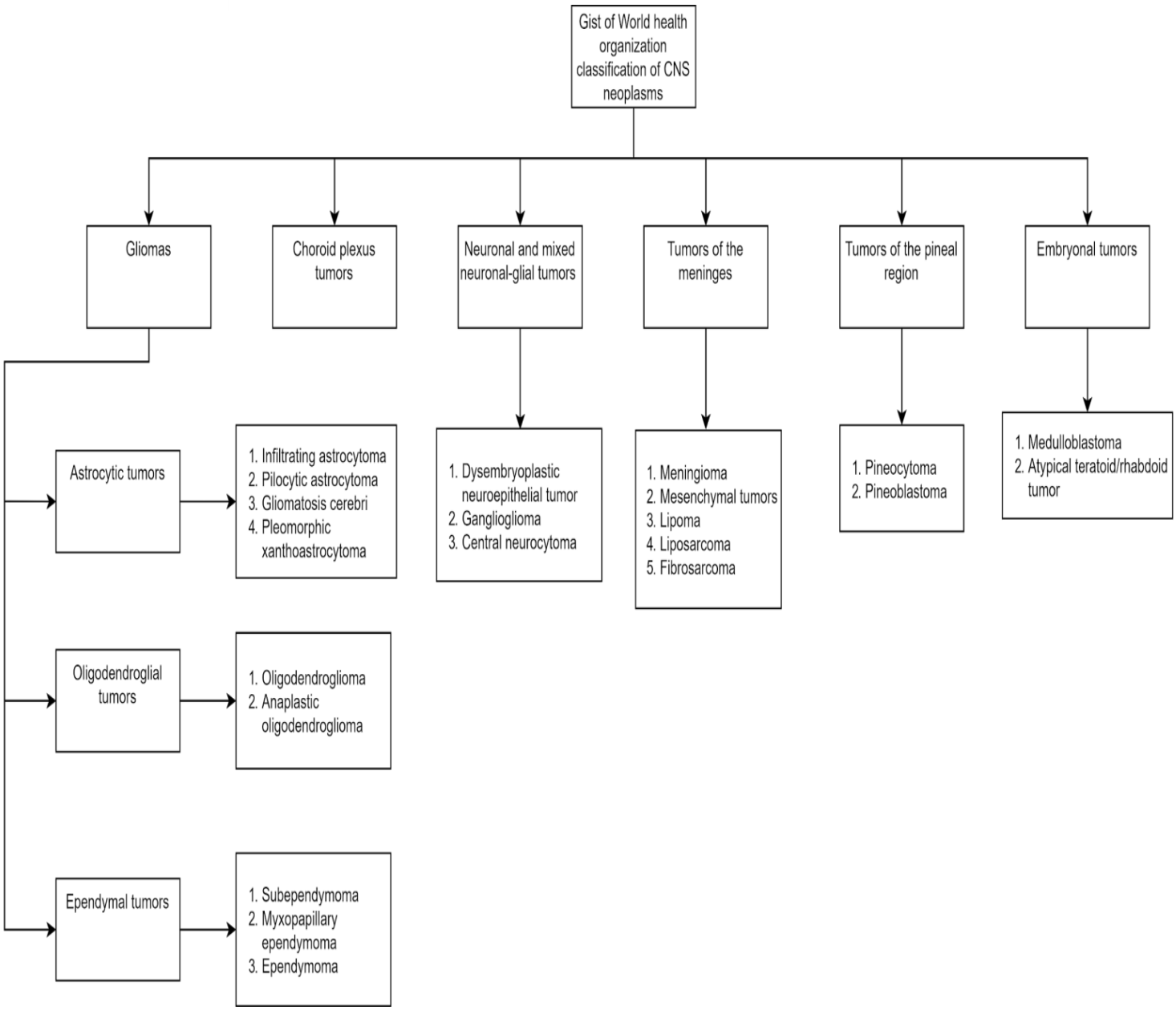
SHORT ANSWERS:

1. Negri bodies
2. Flexner Winsteiner rosettes and fluerretes

UPDATES

ESSAY:

1. TYPES OF TUMORS



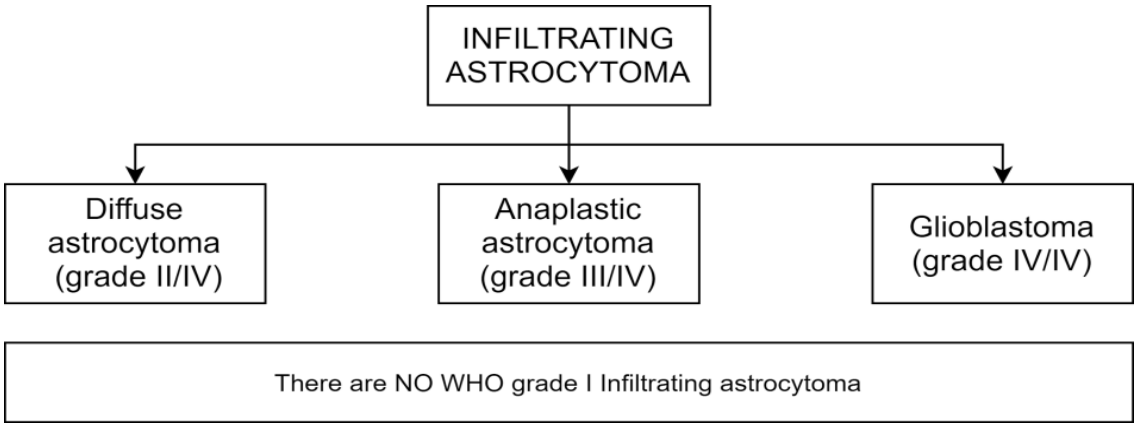
GLIOMAS

- Most common group of primary tumors.
- The classification is based on histologic features.
- They include
  - **Astrocytoma**
  - **Oligodendroglioma**
  - **Ependyma.**

1. ASTROCYTIC TUMOURS (ASTROCYTOMA)

The two major categories are diffusely infiltrating astrocytoma and more localized astrocytoma, of which the most common are **Pilocytic astrocytoma**.

A. INFILTRATING ASTROCYTOMA:



- 80% of the primary brain tumor in adults.
- Commonest in cerebrum, cerebellum and spinal cord.
- Peak during fourth to sixth decade.

Molecular genetics

- It occur in one of two clinical settings.
- Most commonly in older individuals as primary glioblastoma; less frequently in young patients as secondary glioma.
- Based on molecular alteration in glioblastoma, they are classified into four molecular subtypes-

	CLASSICAL	PRO NEURAL	MESENCHYMAL	NEURAL
Associated genomic (DNA) changes	● Amplification of **EGRF oncogene** ● **PTEN tumor** suppressor gene mutation ● Deletion **of ch 10** ● Focal deletion in **ch9p21(del. of CDKN 2a** tumor suppressor gene)	● Point mutation in **IDH1** and **IDH2** ● **PDGFRA** amplification ● Mutation of TP53	● Del. of **NF1 gene on ch 17** ● ↓NF1 protein ● Over-expression of genes in TNF pathway and NF-kB pathway	Higher levels of neuronal markers : ● NEFL ● GABRA1 ● SYT1 ● SLC12A5

	CLASSICAL	PRO NEURAL	MESENCHYMAL	NEURAL
Response to aggressive chemotherapy	Responsive	No survival advantage	Responsive	Possibly responsive
Clinical-pathologic correlation	Primary Glioblastoma	Younger patients enriched for **secondary glioblastoma**	Primary Glioblastoma (mesenchymal)	Primary Glioblastoma

i. DIFFUSE ASTROCYTOMA

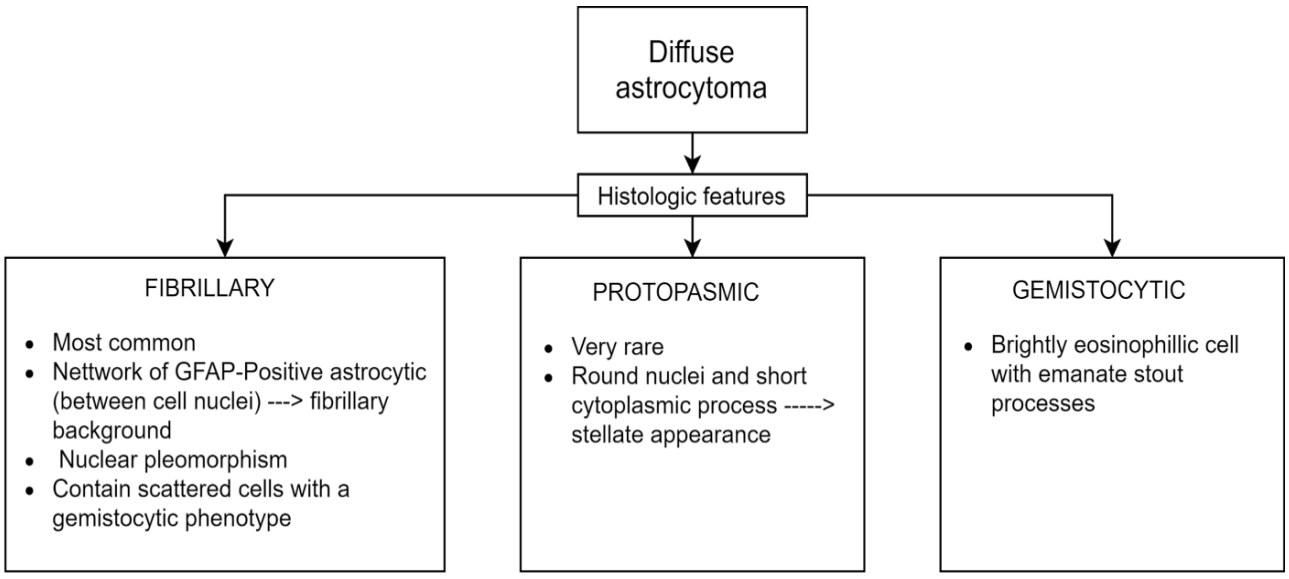
Clinical features

- Epilepsy
- Produce neurologic deficits and symptoms and signs of raised ICP at a later stage
- Acquire areas of contrast enhancement on imaging due to “leaky” blood brain barrier in high-grade
- Variable prognosis

Gross morphology

- Poorly demarcated, gray, infiltrative tumor
- Size : Few cm to enormous lesion replacing hemisphere
- Cut surface : Firm or soft/gelatinous
- Macroscopic cyst formation can occur with clear yellow fluid

Histology



ii. ANAPLASTIC ASTROCYTOMA (III/IV)

Clinical features

- Rapidly worsening neurologic sign
- Conversion of a previously non-enhancing tumor to one enhancing in MRI
- Poor prognosis

Gross morphology

- Poorly differentiated from diffuse astrocytoma
- Distinct mass may be seen due to focal increase in cell proliferation

Histology

- Densely cellular with greater nuclear pleomorphism
- Mitotic activity (differentiates from diffuse astrocytoma)
- Necrosis is not found

iii. GLIOBLASTOMA / GBM / GLIOBLASTOMA MULTIFORME (IV/V)

- **Primary glioblastoma:** New onset typically in old age group **(EFGR over expression)**
- **Secondary glioblastoma**: Progression of low grade astrocytoma typically in younger patients **(IDH1 mutations)**

Gross morphology

- Characteristic variation in the appearance of tumor from region to region
- Firm & white other are soft& yellow due to necrosis
- Cystic degeneration &hemorrhage can also be noted
- In MRI : Spherical mass with a necrotic center (filled with straw coloured fluid with necrotic debris) forming a ring enhancement
- Forms Butterfly glioma if it crosses corpus callosum forming a bi-hemispheric mass.

Histology

- Similar to anaplastic astrocytoma
- **Necrosis :** Serpentine pattern in areas of hypercellularity
- **Pseudo-palisading :** Tumor cells collect along the edges of necrotic region
- Vascular cell proliferation : Cells pile up and budge into the lumen (minimum two layers of endothelial cells)
    - Produces Glomeruloid body
    - Hypoxia induced VEGF production contributes to the changes.

Clinical features

- Very poor prognosis
- Presence of IDH1 mutation (R132H mutation) is associated with better outcome
- Epigenetic silencing of the promoter for the gene encoding DNA repair enzyme MGMT predicts the responsiveness to DNA alkylating agents

B. GLIOMATOSIS CEREBRI

- Diffuse glioma with extensive infiltration of multiple regions of brain
- Grade III/IV

Clinical features

- Symptoms depend on location and growth rate of tumor
- Mean survival rate is more than 5 years

Radiological features

- Mass effect as well as changes in the brain adjacent to the tumour such as edema.
- High grade astrocytoma have abnormal blood vessels that are" leaky '' with an abnormally permeable blood brain barrier.

C. PILOCYTIC ASTROCYTOMA

- It has relatively benign behavior
- Occur in children and young adults
- Juvenile pilocytic astrocytoma: most common benign tumor in children
- Anatomical location : Cerebellum, floor and walls of 3rd ventricle, optic nerve rarely in cerebral hemisphere
- Significantly better prognosis

Gross morphology:

- Cystic or if solid well circumscribed(less frequently infiltrative)
- Symptomatic recurrence of incompletely resected lesions is mostly due to cyst enlargemen

Histology:

- Rosenthal fibres and eosinophilic granules
- Biphasic appearance :
  - Fibrillary meshwork : Formed by bipolar cells with GFAP- positive hair like processes
  - Microcystic component : Stellate cells with branching cytoplasmic processes enclose a fine meshwork of microcysts
- Vascular proliferation does not imply poor prognosis
- Mitotic activity and necrosis is uncommon

Molecular genetics:

- Alteration in BRAF signaling pathway :
  - Translocation which separates the kinase domain from the inhibitory domain
  - Activating point mutation (V600E)
- Of patients with neurofibromatosis type 1, 15% develop a Pilocytic astrocytoma with loss of function of neurofibromin.
- Lack either IDH1 / 2 or TP53 mutations

D. PLEOMORPHIC XANTHOASTROCYTOMA

- Low grade tumor (II/IV) with 5 year survival rate
- Occurs in children and young adults
- Anatomical location : Temporal lobe

Histology:

- Shows considerable polymorphism
- Cytoplasmic lipid accumulation is found in astrocytes
- Extreme nuclear atypia signifies higher grade
- Presence of reticulin deposits, relative circumscription, chronic inflammatory cell infiltrates and absence of mitotic activity and necrosis signify lower grades

2. OLIGODENDROGLIOMA

- **Infiltrating gliomas** comprised of cells resembling oligodendrocytes
- Most common in **4th or 5th decade**
- **Anatomical location** : Cerebral hemisphere (predominately white matter);**frontal lobe**

Molecular Genetics:

- Most common : **IDH1 & IDH2** mutation (better prognosis)
- **Codeletion of 1p/19q** confer consistent responses to chemotherapy and radiation comparatively
- Progression to Anaplastic Oligodendroglioma:
  - Loss of 9p
  - Loss of 10q
  - Mutation in CDKN2

Gross Morphology:

- Well circumscribed, gelatinous gray masses
- Often present with cyst, focal hemorrhage and calcification

Histology:

- Sheets of regular cells with **spherical nuclei with fine granular chromatin** surrounded by **clear halo** of vacuolated cytoplasm
- **Fired egg appearance** of anastamosing capillaries
- Calcification : Microscopic foci to massive deposits
- Mitotic activity minimal or absent
- **Anaplastic Oligodendroglioma:**
  - Higher cell density
  - Nuclear anaplasia
  - Mitotic activity and necrosis

Clinical Features:

- Better prognosis
- Anaplastic oligodendroglioma has worse prognosis
- Takes about 6 years for progression to anaplastic

3. EPENDYMOMA

A. EPENDYMOMA

Arise next to **Ependyma**- lined ventricular system including the oft-obliterated central canal of spinal cord

- **Anatomical location :**
  - In 1st and 2nd decade of life - **Near 4rth ventricle**
  - In adults **(neurofibromatosis type 2 patients) : Spinal cord**
- **Molecular Genetics:**
  - Mutation of **NF2 gene on ch 22**
  - Lack TP53 mutations
  - Two subtypes
    - ✓ **Mesenchymal phenotype expressing:** Increased propensity to metastases and common in younger patients
    - ✓ **Aberrations of large regions of chromosomes:** Better prognosis
- **Gross Morphology:**
  - Solid or papillary masses
  - Well demarcated but complete resection is not possible due to proximity of medullary centres
  - Intraspinal tumors can be completely resected
- **Histology:**
  - Composed of cells with regular round nuclei with granular chromatin
  - Dense fibrillar cytoplasm
  - Tumor cells form gland like elongated structures
  - **Perivascular pseudorosettes:** Tumor cells arranged around vessels with an intervening zone consisting of thin ependymal processes directed towards wall of vessel
  - **Rosettes are seen**
  - **Anaplastic ependymoma:** high cellular density, mitotic activity and necrosis
- **Clinical Features:**
  - Posterior fossa ependymomas present with hydrocephalus due to obstruction of 4rth ventricle and have worst outcome
  - Supratentorial and spinal ependymoma resection is easier and thus better outcome

B. MYXOPAPILLARY EPENDYMOMA

- **Anatomical location : Filum terminale** of the spinal cord
- Prognosis depends on complete surgical resection
- Infiltration into subarachnoid space : recurrence is likely
- **Histology:**
  - Papillary element in **myxoid background with ependymoma like cells**
  - **Myxoid area:** neutral and acidic mucopolysaccharides
  - Cuboidal cells with clear background and connective tissue,
  - Blood vessels are present around the papillary core

C. SUBEPENDYMOMA

- Solid, calcified slow growing nodules attached to ventricular lining and protruding into the ventricle in lateral and fourth
- Asymptomatic
- If large or strategically placed cause hydrocephalus
- **Histology:**
  - Clusters of ependymal - appearing nuclei scattered in a dense, fine glial background

D. CHOROID PLEXUS TUMOUR:

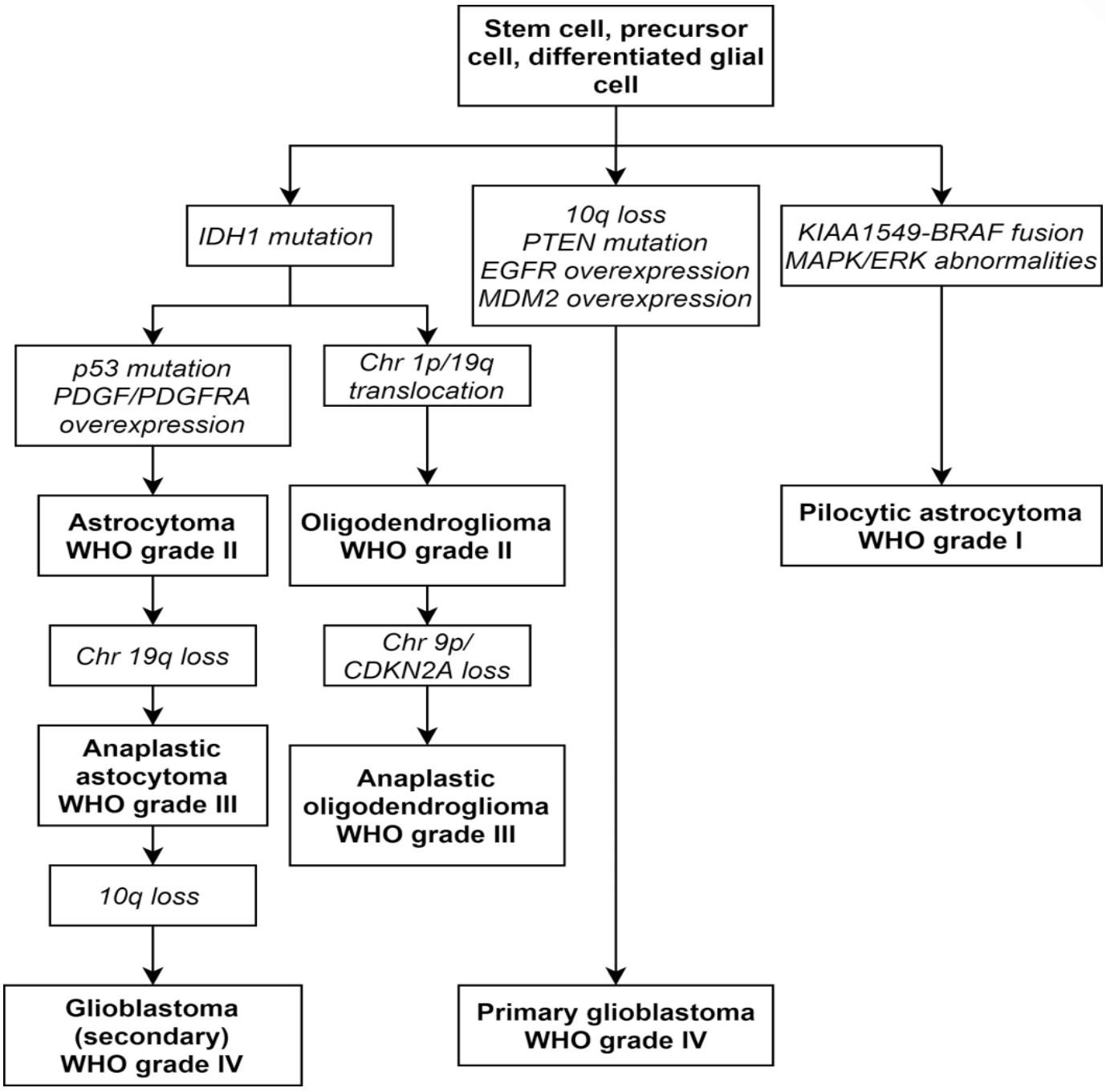
Choroid plexus papilloma:

- Occur any along the choroid plexus.
- Most common in children, usually found in lateral ventricals.
- In adults, it more often involve fourth ventricle.
- The papillae have connective tissue stalks covered with a cuboidal / columnar epithelium.
- **Clinical features:**
  - Usually present with hydrocephalus due to obstruction of the ventricular system by tumor or overproduction of CSF.

Choroid plexus carcinoma:

- Resemble adenocarcinoma
- Usually found in children and adults.

GENETIC PATHWAYS TO FORMATION OF GLIOMAS

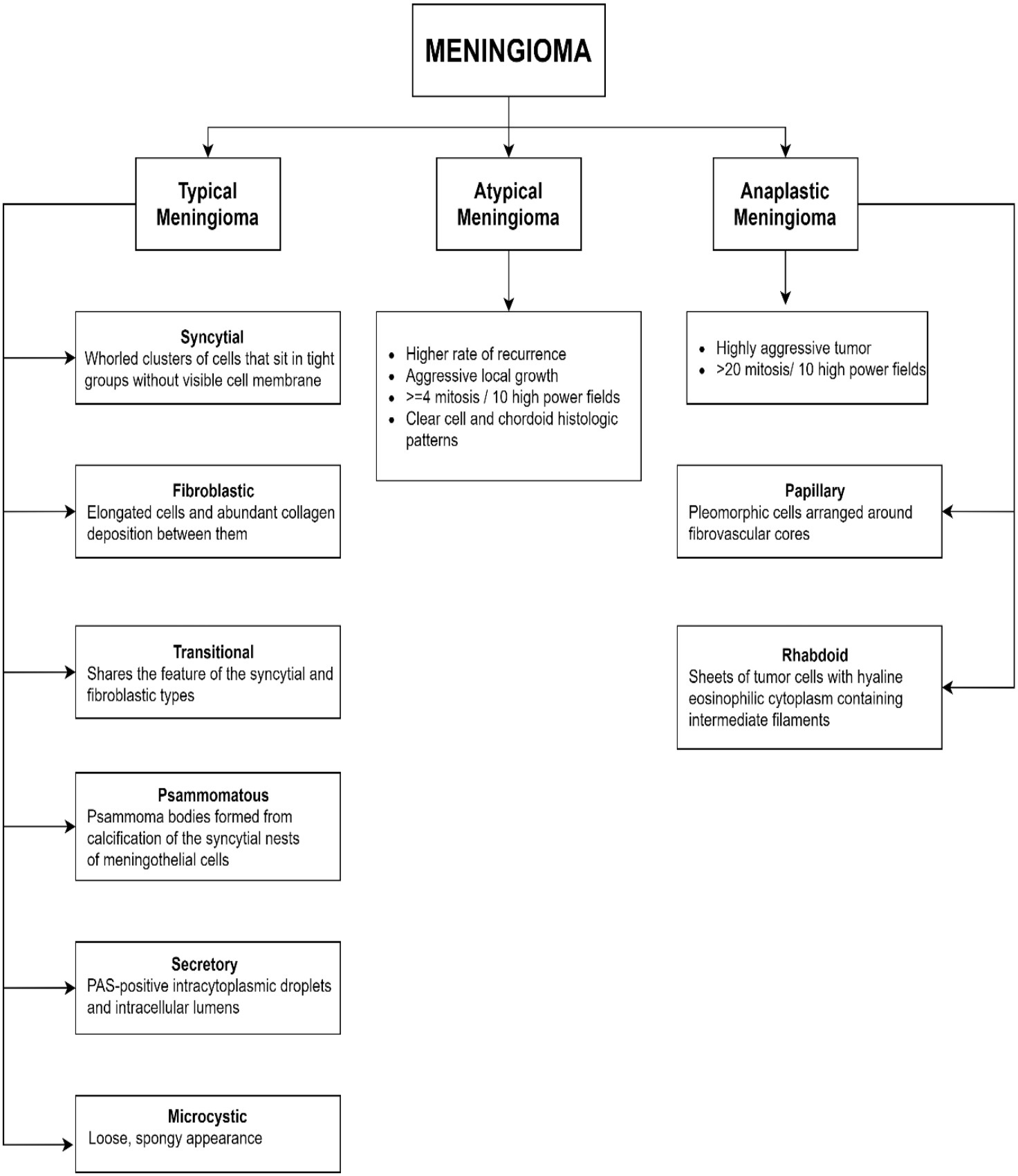


SHORT NOTES

1. MENINGIOMA

- **Benign tumor attached to the dura arising from meningothelial cells of the arachnoid.**
- Most common benign tumor in adults. The incidence of meningiomas rises with age.
- Most meningiomas have low risk of recurrence or aggressive growth
- **Anatomical location:** External surfaces of the brain and within the ventricular system
  - Parasagittal aspect of the brain convexity
  - Dura over the lateral convexity
  - Wing of the sphenoid
  - Olfactory groove
  - Sellaturcica
  - Foramen magnum
- **Female: Male gender bias**
  - 3:2 for intracranial meningiomas.
  - 10:1 for intraspinal meningioma
  - Anaplastic meningiomas affect males and females equally
- **Associations:**
  - Previous radiotherapy.
  - **Estrogen-dependent neoplasm** (breast carcinoma and endometrial carcinoma).
  - **Neurofibromatosis type 2.**
  - **Castle man syndrome** (for choroid and lymphoplasmacyte-rich meningiomas).
  - Meningiomas express progesterone receptor and grow more rapidly during pregnancy
- **Symptoms:**
  - Vague non-localizing symptoms or with focal findings referable to compression of underlying brain
- **Gross morphology:**
  - Rounded masses with defined dural bases
  - Extension into the overlying bone may be present
  - Encapsulated by thin fibrous tissue producing bosselated or polyploid appearance
  - **En plaque form :** Tumor spreads in a sheet like fashion along the surface of the dura Produces hyperstatic reactive changes in adjacent bone
  - Evident necrosis and extent hemorrhage are absent

- **Histology:**
  - Meningiomas are commonly Immunoreactive for epithelial membrane antigen and positive for carcinoembryonic antigen.



2. TUBERCULOSIS MENINGITIS

Tuberculosis of CNS may be due to active disease from other part or an individual lesion.

ETIOLOGY: Tubercle bacilli

MORPHOLOGY:

- Most common pattern is Diffuse meningoencephalitis, subarachnoid space –Greenish, gelatinous or fibrous exudates involves the base of the brain, cisterns, encapsulating nerves.

CSF EXAMINATION:

- Clear and colorless forms cobweb on standing
- CSF shows pleocytosis made up of mixed inflammatory infiltrates containing lymphocytes, plasma cells and macrophages
- Proteins increased
- Sugar decreased
- Chloride decreased

MICROSCOPIC EXAMINATION:

- Florid cases show well-formed granulomas with caseous necrosis and giant cells, subarachnoid space shows obliterative endarteritis and intimal thickening.
- LONG STANDING: a dense fibrous adhesive arachnoiditis mostly around the base of the brain. Hydrocephalous may result.
- CNS involvement may take from one or more well circumscribed intraparenchymal masses (tuberculomas), associated with meningitis.
- The lesion has central area of caseous necrosis surrounded by granulomas, calcification may occur in inactive lesion.

CLINICAL FEATURE:

- Presents with headache, malaise, mental confusion and vomiting.
- Serious complication of chronic tuberculous meningitis is arachnoids' fibrosis producing hydrocephalous and endarteritis producing arterial occlusion and infarction of underlying brain.
- CNS tuberculosis in AIDS is similar but less host reaction
- HIV patients are at a risk of developing mycobacterium avium intracellular- sheets of macrophages filled with organisms with few or no granulation.

3. EXAMINATION OF CSF:

CSF examination is an important part of neurologic evaluation in **non-neoplastic and neoplastic** diseases of CNS.

NORMAL COMPOSITION:

- Appearance : clear and colourless
- Total volume : 120-150ml in adults, 10-60ml in neonates
- Specific gravity: 1.006-1.008
- Sugar: 50-80 mg/dl
- Proteins: 15-45mg/dl
- Bacteria: Nil

A. SPECIMEN COLLECTION:

- CSF is obtained by one of the following techniques:
  - Lumbar puncture
  - Cisternal puncture
  - Ventricular puncture
  - Lateral cervical puncture
- 2 ml of CSF is withdrawn. CSF tap is done by lumbar puncture, indications following 4 categories:
  - Meningeal infection
  - Subarachnoid haemorrhage
  - CNS malignant tumours
  - Demyelinating diseases.
- The specimen should be transported to the laboratory immediately and processed within one hour, cellular degradation occurs giving incorrect results delay in examination sample may be refrigerated, expect for microbial culture.

B. MICROSCOPIC EXAMINATION: TLC and DLC in the CSF

TOTAL LEUKOCYTE COUNT: TLC in CSF manual automated method

Manual method

- WBCs in CSF are quite low. Counting is done in undiluted sample of CSF.
- WBC counting is done:
  - In all 4 peripheral squares Improved Neubauer's chamber
  - In 16 squares in Fuchs-Rosenthal chamber

Automated method

- Electronic particle counter for TLC in blood are not good if the counts are too low.

DIFFERENTIAL LEUKOCYTE COUNT:

- Centrifuge or cytocentrifuge a small volume of CSF → Prepare smears from the sediment → Stain one of the smears with Romanowsky stain → Examine under high power oil immersion of microscope presence of various cells.

- **Increased neutrophils:**
  - Bacterial meningitis
  - Brain abscess
  - Brain infarct
  - Repeated lumber puncture

- **Increased lymphocytes in CSF:**
  - Viral meningitis
  - Tuberculosis meningitis
  - Parasitic meningitis
  - Fungal infections

- **Causing plasma cells in CSF:**
  - Tuberculosis meningitis
  - Syphilitic meningoencephalitis
  - Multiple myeloma
  - Malignant brain tumor

- **Causing lymphocytes and monocytes in CSF:**
  - Viral meningitis
  - Degenerative brain disorders
  - Tuberculosis meningitis
  - Fungal meningitis
  - Sarcoidosis of meningitis

- **Causing malignant cells in CSF:**
  - Metastatic cancers
  - Leukemias
  - Lymphomas
  - Medulloblastoma
  - Ependymoma

C. CHEMICAL EXAMINATION:

Table 7.1: CSF findings in health and various types of meningitis.

	Feature	Normal	Acute pyogenic (bacterial meningitis)	Acute lymphocytic (viral meningitis)	Chronic (tuberculosis meningitis)
1.	**Naked eye appearance**	Clear and colourless	Cloudy or frankly purulent	Clear or slightly turbid	Clear of slightly turbid, forms fibrin coagulum on standing
2.	**CSF pressure**	60-150 mm water	Elevated (above 180 mm water)	Elevated (above 250 mm water)	Elevated (above 300 mm water)
3.	**Cells**	0-4 lympho-cytes/ml	1,000-100,000 neutrophils/ml	10-100 mononuclears/ml	100-1000 mononuclears/ml
4.	**Proteins**	15-45 mg/dl	Raised	Raised	Raised
5.	**Glucose**	50-80 mg/dl	Reduced (usually less than 40 mg/dl)	Normal	Reduced (usually less than 45 mg/dl)
6.	**Bacteriology**	Sterile	Causative organisms present	Sterile	Tubercle bacilli present

D. MICROBIOLOGY EXAMMINATION:

- CSF stained with gram stain
  - Ziehl-Neelsen stain for AFB
  - Indian Ink for capsule for Cryptococcus
- CSF can be subjected to culture for the following:
  - Bacteria
  - Tubercle bacilli
  - Fungus

E. IMMUNOLOGICAL EXAMINATION:

- CSF required for demonstration of:
  - Viral inclusions by immunostains
  - PCR for viral DNA and tuberculosis
  - ELISA for tuberculosis
  - VDRL for syphilis

4. PRION DISEASE:

- Abnormal forms of a cellular protein.
- Causes rapidly progressive neurodegenerative disorders.
- Sporadic, familial or transmitted.
- **Diseases**:
  - **Creutzfeldt-jakob disease**
  - **Gerstmann-straussler-scheinker syndrome**
  - **Fatal familial insomnia**
  - **Kuru**
  - **Scrapie** in sheep and goats.
  - **Mink-transmissible enchephalopathy, chronic wasting disease** in deer and elk.
  - **Bovine spongiform encephalopathy**
- Abnormal form of specific protein termed as prion protein (PrP) are present.
- **Morphology:**
  - Spongiform change
  - Intracellular vacuoles in neurons and glia.
- **Clinical:** Rapidly progressive dementia

PATHOGENESIS AND MOLECULAR GENETICS:

- Degenerative disorder caused by spreading of pathologic protein
- PrP undergoes a conformational change from its normal α-helix protein to abnormal β-pleated sheets isoform
- PrP acquires resistant to digestion with protease such as proteinase k
- **Diagnosis**: Western blotting of tissue extract after partial digestion with proteinase K allows detection of PrP.
- The property of the abnormal protein conformation to induce similar structural changes in other tissue shows its self-propagating property and provides a link between Alzheimer's disease and Parkinson disease.
- Gene encoding PrP is PRNP, shows variety of mutation which are responsible for prions disease
- Polymorphism at codon 129 that encodes either methionine(Met) and valine (Val)
- Heterozygosity at codon 129 is protective against development of the disease the same is observed in iatrogenic CJD

CREUTZFELDT-JAKOB DISEASE (CJD)

- Most common Prion disease
- Familial form- mutation in PRNP
- **Iatrogenic transmission:**
  - Corneal transplantation
  - Deep implantation of electrodes in brain
  - Administration of contaminated preparations of naturally derived growth hormone
- **Morphology:**
  - Spongiform neurons
  - Severe neuronal loss, reactive gliosis
  - Status spongiosus
  - Inflammation absent
  - Kuru plaques are extracellular aggregates of abnormal PrP proteins which are Congo red and PAS positive.
- **Clinical changes:**
  - Rapidly progressive dementia
  - Startle myoclonus
  - Signs of cerebellar dysfunctions- ataxia

FATAL FAMILIAL INSOMNIA

- Sleep disturbance in initial stages
- Caused by mutation in PRNP gene
- Mutation that leads to aspartate substitution for aspargine residue at 178 of PrP
- Neurologic signs- ataxia, autonomic disturbances , stupor and coma
- **Morphology**
  - Neuronal loss
  - Reactive gilosis in anterior ventral, dorsomedial nuclei of thalamus and inferior olivary nuclei
  - Proteinase K resistant PrP– detected by Immunohistochemical assay and western blotting
  - It does not show spongiform pathology

VARIANT CREUTZFELDT-JAKOB DISEASE:

- The disease affected young adults, behavioral disorders in early stages and neurologic syndrome occurs
- The neuropathologic findings and molecular features are similar to CJD
- Pathologic variant CJD characterized by presence of extensive cortical plaques surrounded by halo of spongiform change
- No alteration in the PRNP gene
- vCJD linked to consumption of bovine spongiform encephalopathy agent in contaminated foods or blood transfusions.
- **Morphology:**
  - Evident brain atrophy
  - Spongiform transformation of cerebral cortex,and often deep gray matter structures
  - Advanced cases: severe neuronal loss, reactive gliosis and expansion of the vacuolated into cyst like spaces "status spongiosis"
  - Inflammation absent
- **Electron microscopy:** vacuoles membrane bound and located within cytoplasm of neuronal processes
- **Kuru plaques**- extracellular deposits of aggregated protein, Congo red and PAS positive and occur in cerebellum in vCJD but found in cerebral cortex in case in vCJD.
- **Immunohistochemical staining**- proteinase K resistant PrP

5. ACUTE PYOGENIC INFECTION:

Causative agent:

- In neonates - E.coli and group B streptococci
- In infants- Haemophilus influenza
- In adolescents and in young adults - Neisseria meningitidis
- In extreme of life - Streptococcus pneumoniae and Listeria monocytogenes

Clinical feature of acute pyogenic infection:

- Meningeal irritation and neurologic impairment –headache, photophobia, irritability, clouding of consciousness and neck stiffness

CSF examination:

- Spinal tap yield cloudy or frankly purulent CSF
- Increased pressure
- 90000 neutrophils per cubic millimeter
- Increased protein concentration
- Reduce glucose

6. MEDULLOBLASTOMA

- Highly malignant tumor in children
- CNS tumors composed of primitive undifferentiated cells include:
  - Medulloblastoma
  - Neuroblastoma
  - Retinoblastoma

MEDULLOBLASTOMA:

- Medulloblastoma is the most common variety of primitive neuro-ectodermal tumor (PNET)
- Comprises of 25% of all childhood brain tumors but a quarter of cases occur in patients over the age of 20 years
- **Location in children:** Exclusively cerebellum
- **Site:** roof of fourth ventricle, most common in the midline of cerebellum in the vermis (70%)
- **Most common site in adults:** lateral cerebellar hemispheres

GROSS APPEARANCE:

- The tumor protrudes into the fourth ventricle as a soft, grey-white mass or invades the surface of cerebellum.

MICROSCOPIC EXAMINATION:

- Medulloblastoma is composed of small, poorly-differentiated, round blue cells
- The cells have an ill-defined cytoplasmic processes and has a tendency to be arranged around blood vessels and occasionally forms pseudo rosettes (**Homer-Wright rosettes**).
- Another characteristic of the tumor is its differentiation into glial or neuronal elements

SHORT ANSWERS:

1. NEGRI BODIES:

- Pathognomonic microscopic finding in **RABIES** infection.
- Size- 2-10 micrometers
- Cytoplasmic, round to oval, eosinophilic inclusions.
- Found in pyramidal neurons within Ammon's horn of the hippocampus and Purkinje cells of the cerebellum.
- Rabies virus can be detected within Negri bodies by ultrastructural and immunohistochemical method.

2. FLEXNER-WINTERSTEINER ROSETTES:

- Clusters of columnar cells arranged around central lumens bounded by eosinophilic membranes.
- Analogous to the external membrane of the normal retina.
- Found in retinoblastoma.
- Also seen in pineoblastoma and medulloepithelioma.

FLUERRETES:

- Retinoblastoma cells that have undergone greater photoreceptor differentiation and group together as a bouquet.
- Lack mitosis or necrosis.

UPDATES

1. ONE LINERS

- Wallerian degeneration refers to degeneration of axons after disruption of nerve fibers
- Rather than having a specific pathologic lesion as its correlate, HAND (HIV Associated Neurodegenerative Disease) is most closely related to inflammatory activation of microglial and perivascular macrophages, some of which are infected by HIV
- The IDH-mutant glioblastomas are aggressive but have a better prognosis overall, with an average survival of 2 to 3 years.
- A potential contribution of humoral immunity has also long been suspected, based on the early observation of oligoclonal bands of immunoglobulin in CSF
- In meningiomas without NF2 mutations, the most common mutations occur in TRAF7, KLF4, AKT1, and SMO; most such tumors involve the skull base and have a lower risk of malignant progression.
- Higher-grade meningiomas more often have chromosomal losses involving several chromosomes, TERT-promoter mutations, and homozygous CDKN2A gene deletions.
- Subependymal giant cell astrocytoma is histologically similar to Tuberous Sclerosis Complex (TSC), but larger; it often presents with obstructive hydrocephalus.
- Retinitis pigmentosa is linked to mutations affecting RHO, USH2A, RGPR, and EYS genes
- Unlike cutaneous and conjunctival melanomas, BRAF mutations do not appear to play a role in the pathogenesis of uveal melanomas
- Germline mutations in BAP1 may predispose patients to the development of uveal melanomas, renal cell carcinomas, and malignant mesothelioma
- Conjunctival Intraepithelial Neoplasia is also called as ocular surface squamous neoplasia.
- BRAF V600 mutations may be identified in nearly 40% of conjunctival melanomas.
- Precursor lesion of Conjunctival melanoma is termed primary acquired melanosis with atypia or conjunctival melanocytic intraepithelial neoplasia (C-MIN).

2. C. GATTII INFECTION

- Most cases of cryptococcal infection in immunosuppressed individuals are caused by C. neoformans.
- Recently, a second species, C. gattii, has been recognized and appears more likely to cause disease in immunocompetent individuals.
- C. gattii involvement of the CNS is more likely to take the form of mass lesions (cryptococcomas) and to present with symptoms related to increased CNS pressure due to mass effects.

3. SUBEPENDYMOMA

- Subependymomas (WHO grade I) are solid, sometimes calcified, slow-growing nodules that are usually found protruding into the lateral or fourth ventricle by growing under the ependyma.
- They are usually asymptomatic and are incidental findings at autopsy or imaging.
- However, if they are sufficiently large or strategically located, they can cause obstructive hydrocephalus.
- Subependymomas have a characteristic microscopic appearance, consisting of clusters of ependymal-appearing nuclei scattered in a dense fibrillar background of glial processes.

4. ENVIRONMENTAL FACTORS IN MS

- There is geographic variation in the prevalence of MS, with a higher number of cases diagnosed away from the equator
- It has been proposed that this latitude dependence may be related to a low level of vitamin D (an immune system modulator) in people who are not exposed to sunlight during winter.

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